INPUT: Observed data and model specification

Observed data

j : group membership

Y_{ij} : outcome values

X_{ij} : predictor values

Model specification

Random intercept model

$$Y_{ij} = \gamma_{00} + \gamma_{10}\big(X_{ij} - \bar{X}_{.j}\big) + \gamma_{01}\bar{X}_{.j} + u_{0j} + e_{ij}$$

with distributional assumptions

Parameter estimation

Method: ML / REML

Fixed effects:
$\hat{\boldsymbol{\gamma}} = (\hat{\gamma}_{00},\ \hat{\gamma}_{01},\ \hat{\gamma}_{10})^T$

Variance components:
$\hat{\boldsymbol{\theta}} = \big(\hat{\sigma}^2,\ \hat{\tau}_0^2\big)^T$

Sampling variance derivation

"Intermediate steps"

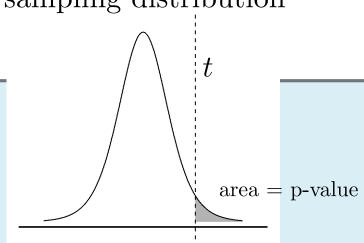
Model-implied covariance of Y

$$\hat{V} = \hat{\tau}_0^2 Z Z^\top + \hat{\sigma}^2 I$$

Standard errors of $\hat{\gamma}$

$$SE(\hat{\gamma}_k) = \sqrt{\Big[(X^\top \hat{V}^{-1} X)^{-1}\Big]_{kk}}$$

sampling distribution



Statistical Inference

Test statistic: $t = \dfrac{\hat{\gamma}_k}{SE(\hat{\gamma}_k)}$

Sampling distribution:
$\mathcal{N}(0,1)$ (large samples) or t_{df} with appropriate degrees of freedom

$\Rightarrow$ **p-values and confidence intervals for $\hat{\gamma}_k$**

Illustrative example. For a hypothetical data set with four persons nested in two groups:

Illustrative observed values

id	j	X_{ij}	Y_{ij}
1	A	2	5
2	A	1	4
3	B	3	2
4	B	5	5

The corresponding model matrices are $X = \begin{pmatrix} 1 & 2 \\ 1 & 1 \\ 1 & 3 \\ 1 & 5 \end{pmatrix}$ $Z = \begin{pmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{pmatrix}$ $Y = \begin{pmatrix} 5 \\ 4 \\ 2 \\ 5 \end{pmatrix}$

and the implied covariance matrix is $\hat{V} = \begin{pmatrix} \hat{\tau}_0^2 + \hat{\sigma}^2 & \hat{\tau}_0^2 & 0 & 0 \\ \hat{\tau}_0^2 & \hat{\tau}_0^2 + \hat{\sigma}^2 & 0 & 0 \\ 0 & 0 & \hat{\tau}_0^2 + \hat{\sigma}^2 & \hat{\tau}_0^2 \\ 0 & 0 & \hat{\tau}_0^2 & \hat{\tau}_0^2 + \hat{\sigma}^2 \end{pmatrix}$